# TESSERA

- It takes raw satellite data such as sentinel-1 and sentinel-2 for an entire year.
- It converts them into tessera embeddings for every pixel (10m X 10m)
- By embedding I mean it creates a compact numerical representation.
- It works as a smart fingerprint for that particular location

What if data are missing?

- Tessera uses self supervised machine learning technique which takes multiple views of the same location and then trains itself. Even if some data are missing in a year, it is highly likely to produce the same embeddings. The data may be missing due to –
    - one month may have many cloudy days
    - another may have few satellite passes
    - some observations may be missing entirely

**Barlow Twins**

This is a machine learning technique that encourages:

- similar inputs → similar embeddings,
- but avoids collapsing everything into identical outputs.

In plain English:
it learns meaningful similarities without becoming too generic.

## What is a Foundation Model?

A **foundation model** is a large pretrained AI model that learns general patterns from huge datasets and can later be adapted for many tasks.

That embedding captures:

- seasonal vegetation behavior,
- moisture,
- land-cover type,
- texture,
- radar response,
- temporal patterns,
- environmental characteristics.

And produces the numerical embedding something like [0.12, -1.7, 0.44, ...] for every pixel.

Tessera is something what LLM’s are for texts, but for the earth observations.